

Extended sign-kill for $\widehat{\lambda} = (2, 2)$: the complete E_j^- chain at all $q \geq 2$

Clio

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Abstract

Let $\lambda = (2, 2, 1^q) \vdash n = q + 4$ with $q \geq 2$, $\ell = q + 2$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda := R'_{\ell+1} \cdots R'_n V_\lambda$. We prove the extended sign-kill conjecture for this family:

$$B \subseteq \bigcap_{j=1}^{q-1} E_j^- |_{V_\lambda} \quad \text{and} \quad B \not\subseteq E_j^- |_{V_\lambda} \quad \text{for } j \geq q.$$

Combined with the universal $B \subseteq E_\ell^+$ (Theorem A, May-13 evening), this completely identifies the T_j -eigenspace containments of B for the family.

The proof is short. The May-13 paper `2026-05-13-non-hook-22-1n.tex` established $\dim B = 1$ and identified the explicit 4-SYT support of the spanning vector b . We observe that this support immediately forces the column-1 entries $1, 2, \dots, n-4$ to occupy column-1 rows $1, 2, \dots, n-4$ in every support SYT. The Hoefsmit same-column rule gives $T_j v_T = -v_T$ for each support SYT and $1 \leq j \leq n-5 = q-1$, hence $T_j b = -b$. Sharpness for $j = q, q+1, \dots$ follows from the non-vanishing of the four support coefficients (also from May-13) together with the off-diagonal Hoefsmit blocks at axial distance ≥ 2 .

This closes the $\widehat{\lambda} = (2, 2)$ row of the May-14 extended sign-kill conjecture table, in both directions.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n-1$, set $S_j := T_j + 1$. The Hecke relation factors as $S_j(T_j - q) = 0$, equivalently $T_j S_j = q S_j = S_j T_j$. Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda}.$$

Since the T_j acts with minimal polynomial $(T - q)(T + 1)$ and the two eigenvalues are distinct (for generic q), $V_\lambda = E_j^+ \oplus E_j^-$.

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega_\ell^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n \in H_q, \quad B := \Omega \cdot V_\lambda.$$

For $\lambda = (2, 2, 1^q)$, $n = q + 4$ and $\ell = q + 2$, so $\Omega = R'_{q+3} R'_{q+4}$ has two R' -blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{(1,1)} = 1$ for $q \geq 0$ (stable regime starts at $q = 0$ here).

SYT labelling for $\lambda = (2, 2, 1^q)$. An SYT T of λ has cell $(1, 1)$ always equal to 1 and column 2 consisting of two cells $(1, 2)$ and $(2, 2)$. Setting $a := T(1, 2)$ and $c := T(2, 2)$, we have $2 \leq a < c \leq n$ and $(a, c) \neq (2, 3)$ (the latter forced by row-2 increasing). Conversely, every such pair (a, c) yields a unique SYT, denoted $T^{(a, c)}$, by filling column 1 at rows $1, 2, \dots, n-2$ with the elements of $\{1\} \cup \{2, \dots, n\} \setminus \{a, c\}$ in increasing order. Total count: $\binom{n-1}{2} - 1 = f^\lambda$.

2 Input from prior papers

Theorem 1 (Support of B , May-13). *For $\lambda = (2, 2, 1^q)$ with $q \geq 2$ (i.e. $n \geq 6$),*

$$B = \text{span}(b), \quad b = \sum_{(a, c) \in \mathcal{S}_n} C_{(a, c)} v_{T^{(a, c)}},$$

where

$$\mathcal{S}_n := \{(n-3, n-2), (n-3, n-1), (n-2, n), (n-1, n)\}$$

and all four coefficients $C_{(a, c)} \in \mathbb{Q}(q)$ are nonzero.

Proof. This is precisely Theorem 1 (combined with Proposition 5.4 on the terminal step) of 2026-05-13-non-hook-

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Theorem 2 (Universal T_ℓ scalar, May-13 evening). *For any partition $\lambda \vdash n$ with $1 \leq \ell = \ell(\lambda) \leq n-1$, $T_\ell \cdot \Omega^{(\lambda)} = q \cdot \Omega^{(\lambda)}$ in H_q . In particular, $B \subseteq E_\ell^+|_{V_\lambda}$.*

3 The extended sign-kill theorem

Theorem 3 (Main). *For $\lambda = (2, 2, 1^q)$ with $q \geq 2$, let $\ell = q+2$, $n = q+4$. The image $B = \Omega \cdot V_\lambda$ satisfies*

$$B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-1} E_j^-|_{V_\lambda},$$

and this is sharp:

$$B \not\subseteq E_j^- \text{ for } j \in \{q, q+1, \dots, n-1\}, \quad B \not\subseteq E_j^+ \text{ for } j \in \{1, \dots, n-1\} \setminus \{\ell\}.$$

We prove the four containments and the four non-containments separately.

3.1 E_ℓ^+ containment

Immediate from Theorem 2.

3.2 E_j^- containment for $1 \leq j \leq q-1$

The key combinatorial observation:

Lemma 4 (Column 1 structure of support SYTs). *For $(a, c) \in \mathcal{S}_n$ with $a < c$, column 1 of $T^{(a, c)}$ holds the elements of $\{1, 2, \dots, n\} \setminus \{a, c\}$ at rows $1, 2, \dots, n-2$ in increasing order; in particular $T^{(a, c)}(i, 1) = i$ if and only if $i \leq a-1$.*

Proof. By definition of $T^{(a,c)}$. Since $a < c$, the sorted column-1 sequence is

$$1, 2, \dots, a-1, a+1, a+2, \dots, c-1, c+1, c+2, \dots, n.$$

The i -th element (occupying row i of column 1) equals i iff $i \leq a-1$. □ □

Remark 5 (The three bounds we will use). By Lemma 4, the column-1 “identity prefix” has length $a-1$ in $T^{(a,c)}$. We will repeatedly use:

- All four members of $\mathcal{S}_n$ have $a \geq n-3$, giving $T^{(a,c)}(i, 1) = i$ for $i \leq n-4$.
- $(n-2, n)$ and $(n-1, n)$ have $a \geq n-2$, giving the additional $T^{(a,c)}(n-3, 1) = n-3$.
- $(n-1, n)$ has $a = n-1$, giving also $T^{(a,c)}(n-2, 1) = n-2$.

Proposition 6 (E_j^- containment). *For $1 \leq j \leq q-1$ and every $(a, c) \in \mathcal{S}_n$, $T_j v_{T^{(a,c)}} = -v_{T^{(a,c)}}$. Consequently $T_j b = -b$, i.e., $b \in E_j^-$.*

Proof. Let $T := T^{(a,c)}$. For $1 \leq j \leq q-1 = n-5$, both j and $j+1$ lie in $\{1, 2, \dots, n-4\}$. By Lemma 4, the entries j and $j+1$ are at cells $(j, 1)$ and $(j+1, 1)$ of T ; in particular they occupy the same column (column 1) at adjacent rows. By the Hoefsmit seminormal action (same-column case), $T_j v_T = -v_T$.

By linearity over $(a, c) \in \mathcal{S}_n$ in $b = \sum_{(a,c)} C_{(a,c)} v_{T^{(a,c)}}$ (Theorem 1), $T_j b = -b$. □ □

3.3 E_j^- non-containment for $q \leq j \leq n-1$

The argument uses non-vanishing of the four support coefficients together with the existence of an off-diagonal Hoefsmit block.

Lemma 7 (Off-diagonal action at $j = q$). *For $j = q$ (so $j+1 = q+1$) and the support SYT $T^* := T^{(n-3, n-2)} = T^{(q+1, q+2)}$, the entries q and $q+1$ sit at cells $(q, 1)$ and $(1, 2)$ of T^* respectively (different row and different column). The corresponding axial distance is $d = c(1, 2) - c(q, 1) = 1 - (1 - q) = q$. In particular, the Hoefsmit T_q -action on the 2-block $\{v_{T^*}, v_{s_q T^*}\}$ is non-degenerate over $\mathbb{Q}(q)$, i.e., $T_q v_{T^*} = \alpha v_{T^*} + \beta v_{s_q T^*}$ with $\beta \neq 0$.*

Here $s_q T^ = T^{(q, q+2)}$ (swap entries q and $q+1$, which exchanges $T^*(q, 1) = q$ with $T^*(1, 2) = q+1$); this is a valid SYT of λ and is not an element of $\mathcal{S}_n$.*

Proof. Lemma 4 gives $T^*(q, 1) = q$. By definition of $T^* = T^{(q+1, q+2)}$, $T^*(1, 2) = q+1$. So $j = q$ at $(q, 1)$ (content $1-q$) and $j+1 = q+1$ at $(1, 2)$ (content 1). Different row and column, axial distance $d = 1 - (1 - q) = q$. For $q \geq 2$ ($\Leftrightarrow |d| \geq 2$), the Hoefsmit ratio $\tau(d) = (q^{|d|} - 1)/(q^d \cdot (q-1)) \cdot (\cdot)$ (the precise formula does not matter) gives $\beta \neq 0$ over $\mathbb{Q}(q)$. Concretely, the off-diagonal entry of the standard Hoefsmit 2-block at axial distance d is a nonzero rational function in q for all $|d| \geq 2$.

Swapping q and $q+1$ in T^* yields the SYT $T^{(q, q+2)}$ (with column-2 entries $q, q+2$). Since $q < q+1$ and $(q, q+2) \notin \mathcal{S}_n$ (which only contains pairs with both entries $\geq n-3 = q+1$), this swapped SYT lies outside the support of b . □ □

Proposition 8 ($B \not\subseteq E_q^-$). $T_q b \neq -b$, hence $b \notin E_q^-$.

Proof. By Theorem 1, $b = \sum_{(a,c) \in \mathcal{S}_n} C_{(a,c)} v_{T^{(a,c)}}$ with all four $C_{(a,c)} \neq 0$. Consider the coefficient of $v_{T^{(q, q+2)}}$ in $T_q b$. This SYT is *not* in $\mathcal{S}_n$, so it is not in the support of b ; hence the coefficient of $v_{T^{(q, q+2)}}$ in $-b$ is 0. On the other hand, by Lemma 7, $T_q v_{T^*}$ produces the term $\beta v_{T^{(q, q+2)}}$ with $\beta \neq 0$. We need to check that no other summand of $T_q b$ cancels this contribution.

A summand $T_q(C_{(a,c)} v_{T^{(a,c)}})$ produces a term proportional to $v_{T^{(q, q+2)}}$ only if either

- $T^{(a,c)} = T^{(q,q+2)}$ (impossible, since $T^{(q,q+2)} \notin \mathcal{S}_n$),
- or $s_q T^{(a,c)} = T^{(q,q+2)}$.

$s_q T^{(a,c)}$ swaps the cells of q and $q+1$ in $T^{(a,c)}$. The resulting SYT has the same column-2 entries as $T^{(a,c)}$ except that any occurrence of q or $q+1$ in column 2 is swapped.

Among the four $(a, c) \in \mathcal{S}_n$:

- $(a, c) = (q+1, q+2)$: column 2 entries are $q+1, q+2$, both in column 2. Swap of $q, q+1$ moves $q+1$ from $(1, 2)$ to $(q, 1)$ and q from $(q, 1)$ to $(1, 2)$ (provided $T^{(a,c)}(q, 1) = q$, which Lemma 4 gives). Resulting SYT: column-2 entries $q, q+2$, i.e., $T^{(q,q+2)}$. *This is the only contributor.*
- $(a, c) = (q+1, q+3)$: same as above for the position of $q+1$ at $(1, 2)$. Swap $q \leftrightarrow q+1$: now q at $(1, 2)$, $q+1$ at $(q, 1)$. Column-2 entries: $q, q+3$, i.e., $T^{(q,q+3)}$. *Different from $T^{(q,q+2)}$.*
- $(a, c) = (q+2, q+4)$: here $a = q+2 \geq n-2$, so by Lemma 4 the identity prefix has length $a-1 = q+1$, giving $T^{(a,c)}(q, 1) = q$ and $T^{(a,c)}(q+1, 1) = q+1$. Hence q and $q+1$ are in column 1 at adjacent rows, so $T_q v_{T^{(a,c)}} = -v_{T^{(a,c)}}$ (same-column case). *No off-diagonal contribution.*
- $(a, c) = (q+3, q+4)$: here $a = q+3 = n-1$, identity prefix length $q+2$, so analogously $T_q v_{T^{(a,c)}} = -v_{T^{(a,c)}}$.

Therefore the coefficient of $v_{T^{(q,q+2)}}$ in $T_q b$ equals $C_{(q+1,q+2)} \beta \neq 0$, while the same coefficient in $-b$ is 0. Hence $T_q b \neq -b$, so $b \notin E_q^-$. $\square$

Proposition 9 ($B \not\subseteq E_j^-$ for $q+1 \leq j \leq n-1$). *For each $j \in \{q+1, q+2, \dots, n-1\}$, $T_j b \neq -b$.*

Proof. We treat the three sub-ranges separately.

$j = q+1 = \ell-1 = n-3$. Positions of $j, j+1$ in each support SYT $T^{(a,c)}$:

- $(a, c) = (n-3, n-2)$: $j = a$ at $(1, 2)$, $j+1 = c$ at $(2, 2)$. *Same column:* $T_j v = -v$.
- $(a, c) = (n-3, n-1)$: $j = a$ at $(1, 2)$, $j+1 = n-2$ in column 1 (specifically at row $n-3$). *Off-diagonal*, axial distance $1 - (-(n-4)) = n-3$.
- $(a, c) = (n-2, n)$: $j = n-3$ in column 1 at row $n-3$, $j+1 = a = n-2$ at $(1, 2)$. *Off-diagonal*, axial distance $-(n-3)$.
- $(a, c) = (n-1, n)$: $j, j+1 = n-3, n-2$ both in column 1 at rows $n-3, n-2$. *Same column:* $T_j v = -v$.

Swap of $j, j+1$ in $T^{(n-3,n-1)}$: $j = n-3$ moves from $(1, 2)$ to column 1 row $n-3$; $j+1 = n-2$ moves from column 1 row $n-3$ to $(1, 2)$. Resulting SYT: column 2 entries $(n-2, n-1)$, i.e. $T^{(n-2,n-1)}$, which is outside $\mathcal{S}_n$. Swap of $j, j+1$ in $T^{(n-2,n)}$: analogously yields $T^{(n-3,n)}$, also outside $\mathcal{S}_n$. Since these two swap-targets are distinct from each other and both have $|d| = n-3 \geq 3$ (non-degenerate Hoefsmit block), each contributes a nonzero coefficient to $T_{q+1} b$ on a SYT outside $\mathcal{S}_n$, while $-b$ is supported entirely in $\mathcal{S}_n$. Hence $T_{q+1} b \neq -b$.

$j = \ell = n-2$. By Theorem 2, $T_\ell b = qb$. For generic $q \neq -1$, $qb \neq -b$, so $b \notin E_\ell^-$.

$j = \ell + 1 = n - 1$. Positions of $j, j + 1 = n - 1, n$ in each support SYT:

- $(n - 3, n - 2)$: column 1 entries are $\{1, 2, \dots, n - 4, n - 1, n\}$ placed at rows $1, \dots, n - 2$ in increasing order, so $n - 1$ at row $n - 3$ and n at row $n - 2$ of column 1. *Same column*: $T_{n-1}v = -v$.
- $(n - 3, n - 1)$: column 1 omits $n - 3$ and $n - 1$, so column 1 = $\{1, \dots, n - 4, n - 2, n\}$ at rows $1, \dots, n - 2$. $n - 1$ at $(2, 2)$, n at row $n - 2$ of column 1. Different row, column. Off-diagonal.
- $(n - 2, n)$: column 1 = $\{1, \dots, n - 3, n - 1\}$. $n - 1$ at row $n - 2$ of column 1, $n = c$ at $(2, 2)$. Off-diagonal.
- $(n - 1, n)$: $n - 1, n$ at $(1, 2), (2, 2)$. Same column 2. $T_{n-1}v = -v$.

Swap of $j, j + 1$ in $T^{(n-3, n-1)}$: $n - 1$ moves from $(2, 2)$ to column 1 row $n - 2$; n moves from column 1 row $n - 2$ to $(2, 2)$. Resulting SYT: column 2 entries $(n - 3, n)$, i.e. $T^{(n-3, n)}$, outside $\mathcal{S}_n$. Swap in $T^{(n-2, n)}$: analogously yields $T^{(n-2, n-1)}$, also outside $\mathcal{S}_n$. Same argument as in the $j = q + 1$ case: each off-diagonal contribution gives a nonzero coefficient on a distinct SYT outside the support of b . Hence $T_{n-1}b \neq -b$. $\square$ $\square$

3.4 E_j^+ non-containment for $j \neq \ell$

Proposition 10 ($B \not\subseteq E_j^+$ for $j \neq \ell$). For $j \in \{1, \dots, n - 1\} \setminus \{\ell\}$, $T_j b \neq qb$.

Proof. For $j \leq q - 1$, $T_j b = -b$ by Proposition 6, which equals qb iff $-1 = q$, impossible for generic q . For $j \geq q$, the same off-diagonal arguments of Propositions 8–9 show that $T_j b$ has a nonzero component on an SYT outside $\mathcal{S}_n$; but qb is supported on $\mathcal{S}_n$. Hence $T_j b \neq qb$. $\square$ $\square$

This completes the proof of Theorem 3.

4 Computational verification

Theorem 11 (Verification). *Theorem 3, together with the support description of Theorem 1, was verified computationally at $q = 2, 3, 4, 5, 6, 7$ (i.e., $n = 6, \dots, 11$) mod $P = 100003$ at $q = 11$. Across these six cases:*

- $\dim B = 1$ in all cases.
- Support of b matches $\mathcal{S}_n$ exactly, all four coefficients nonzero.
- $T_j b = -b$ for $1 \leq j \leq q - 1$ in all cases.
- $T_q b \neq -b$ in all cases (verified by recovering a nonzero coefficient on $T^{(q, q+2)}$).
- $T_\ell b = qb$ in all cases (Theorem 2).

Proof. Scripts `~/projects/scratch/2026-05-14-extended-sign-kill-proof/verify_support_argument.py` and `check_S10mega_zero.py`. The scripts construct the Hoefsmit T_i matrices for $\lambda = (2, 2, 1^q)$, build Ω explicitly, and check all of the above by direct linear algebra. $\square$ $\square$

5 Discussion

5.1 What is new

The extended sign-kill conjecture (May-14 evening) asserts, for $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$,

$$B \subseteq E_j^-|_{V_\lambda} \iff j \leq \tau(\lambda),$$

where $\tau(\lambda) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$. For $\hat{\lambda} = (2, 2)$, $\tau = q - 1$.

The May-13 paper `2026-05-13-non-hook-22-1n.tex` proves the $j = 1$ case (containment in E_1^-). The May-20 paper proves the analogous $j = 1$ case for $\hat{\lambda} = (3, 2)$ at $q \geq 3$. Neither paper addresses $j \geq 2$.

The present paper closes the $\hat{\lambda} = (2, 2)$ row of the conjecture in both directions: $B \subseteq E_j^-$ for $j \leq q - 1$, and $B \not\subseteq E_j^-$ for $j \geq q$. The proof is short and extracted directly from the support structure already in May-13; the contribution is the observation that the support immediately forces $T(i, 1) = i$ for $i \leq n - 4$, plus the sharpness argument using the non-vanishing of the four support coefficients.

5.2 Why this is plausible to extend

The argument generalizes mechanically to any shape λ where:

- (1) $\dim B$ is known and the explicit support of a basis of B is known.
- (2) The support SYTs share a common “top-of-column-1” pattern, i.e., there is some i_0 such that every support SYT has $T(i, 1) = i$ for $i \leq i_0$.
- (3) There is a sharpness witness at $j = i_0 + 1$, again following from the support description.

For $\hat{\lambda} = (3, 2)$, the May-20 paper identifies B as the image of the three-branch decomposition under Φ_X ; this gives a candidate support structure that should allow an analogous argument for E_j^- at $j \leq q - 2$. We sketch the extension as future work.

5.3 Containment versus identity

Theorem 3 gives $B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-1} E_j^-$. The intersection on the right is computationally much larger than $\dim B = 1$ (e.g., dimension 2 for $\lambda = (2, 2, 1^q)$ at all tested $q \geq 2$, see May-14 evening paper’s Table 2). So the eigenspace constraints do not pin down B exactly; the closed-form $\dim B = f^{\lambda^{(\geq 2)}} = 1$ is a sharper fact, capturing structure not detected by T_j -eigenspaces alone.

5.4 Status of the broader iso programme

This result closes one row of the May-14 evening conjecture table but does not affect the broader iso programme (the May-13–May-14 attempts to identify B as a known representation-theoretic object). Those attempts were closed off by the May-14 morning and afternoon refutations. The present contribution is, in effect, a “cleaning up” of the extended sign-kill empirical observation into a structurally established theorem at the simplest non-hook shape.

6 Status

Established.

- Theorem 3: complete characterization of T_j - eigenspace containments of B for $\lambda = (2, 2, 1^q)$, $q \geq 2$.
- Sharpness for both E_j^- and E_j^+ containments.

Open.

- Extended sign-kill at $\hat{\lambda} = (3, 2)$ at $j \geq 2$ (requires May-20-style support description; computationally consistent with threshold $\tau = q - 2$).

- Extended sign-kill at general $\widehat{\lambda}$ (requires support description for general multi-corner shapes).
- Structural mechanism for the top-two-rows independence $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ (independent of lower rows of $\widehat{\lambda}$).

Push status. TBD — depends on PAT refresh. Prior unpushed-commit count on `clio-vega/proofs`: 64 (May-14 evening session).

Scripts. All computations: `~/projects/scratch/2026-05-14-extended-sign-kill-proof/`.

- `explore_22_family.py` — ranks and intersections of $E_j^\pm \cap B$ for $\lambda = (2, 2, 1^q)$ at $q = 2, 3, 4$.
- `identify_22.py` — explicit basis vector of B in Hoefsmit basis, support count, normalized coefficients.
- `check_S10mega_zero.py` — verifies $S_1\Omega = 0$ as a matrix on V_λ for $q \geq 2$ in the $(2, 2, 1^*)$ family, and diagnostic checks on other families $((3, 2, 1^*), \text{etc.})$.
- `verify_support_argument.py` — end-to-end verification of Theorem 3: support matches $\mathcal{S}_n$, the column-1 pattern, and the T_j -action at $j \leq q - 1$ and $j = q$.